A project report on

CONCURRENT POST-QUANTUM ACCELERATION ON CONSUMER HARDWARE

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology in Computer Science and Engineering

by

RAVI VENAKTA SAI KARTIK (22BCE1743)

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Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

November, 2025

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DECLARATION

I hereby declare that the thesis entitled "CONCURRENT POST-QUANTUM ACCELERATION ON CONSUMER HARDWARE" submitted by ABHINAV KUSHWAHA (22BCE1746), for the award of the degree of Bachelor of Technology in Computer Science and Engineering, Vellore Institute of Technology, Chennai is a record of bonafide work carried out by me under the supervision of DR. Karmel A.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Chennai

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CERTIFICATE

This is to certify that the report entitled **"CONCURRENT POST-QUANTUM ACCELERATION ON CONSUMER HARDWARE"** is prepared and submitted by **Abhinav Kushwaha (22BCE1746)** to Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

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ABSTRACT

The paper introduces an innovative approach to enhancing the performance of CRYSTALS-Kyber, a post-quantum cryptosystem, by utilizing AI accelerators, specifically NVIDIA's Tensor Core. The authors identify polynomial multiplication as one of the most computationally intensive tasks in lattice-based cryptography, and they aim to accelerate this operation using Tensor Cores, which are traditionally designed for machine learning tasks. The challenge lies in adapting the cryptographic computations to suit the specialized nature of these accelerators, which are optimized for low-precision arithmetic.

To overcome this, the authors transform cryptographic operations into a matrix multiplication format that Tensor Cores can efficiently process. They also balance precision and performance, experimenting with different data types such as INT8 and FP16, to maintain the correctness of the cryptographic operations while achieving significant performance gains. The resulting implementation shows impressive speedups—about 6.47 times faster than the state-of-the-art on the same hardware. The paper further introduces an "NTT box" that performs the Number Theoretic Transform (NTT) and its inverse (INTT) more efficiently, using precomputed twiddle factor tables to reduce memory access delays.

This implementation is tested on an RTX 3080 GPU and achieves substantial improvements in speed compared to previous CPU and GPU-based implementations, demonstrating the potential of AI accelerators in cryptographic computing. The authors also address security concerns, ensuring that the use of Tensor Cores does not introduce new vulnerabilities and may, in fact, enhance resistance to side-channel attacks. Overall, this work presents a promising framework for using AI accelerators to accelerate post-quantum cryptography, offering a path forward for more efficient cryptographic algorithms in the post-quantum era.

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Place: Chennai

Date: **Abhinav Kushwaha**

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LIST OF ACRONYMS

1. CUDA: Compute Unified Device Architecture
2. GUI: Graphical User Interface
3. IMGUI: Immediate Mode GUI
4. IPC: Instruction Per Cycle
5. WMMA: Warp Matrix Multiply and Accumulate

Chapter 1
Introduction, Objective and Literature Review

1.1 INTRODUCTION

Public-key cryptography is a core building block of modern secure communication. Schemes such as RSA and elliptic-curve cryptography are widely deployed because they provide strong security with high performance on commodity hardware. However, the emergence of large-scale quantum computers poses a direct threat to these systems. Shor's algorithm shows that integer factorization and discrete logarithms can be solved efficiently on a quantum computer, which would render today's public-key schemes insecure.

To address this threat, the cryptographic community has shifted attention toward post-quantum cryptography (PQC), which is designed to remain secure against both classical and quantum attacks. As part of this effort, the NIST Post-Quantum Cryptography Standardization process selected CRYSTALS-Kyber as the standard public-key encryption and key-establishment algorithm. Kyber is based on lattice assumptions, specifically the Module Learning With Errors (MLWE) problem, which is believed to be resistant to known quantum attacks.

As a result, hardware acceleration is widely seen as essential for making post-quantum cryptography practical at scale. GPUs are particularly attractive for this task due to their massive parallelism and high arithmetic throughput. However, achieving good performance on GPUs is not straightforward. Naïve implementations suffer from excessive global memory traffic, poor data locality, and underutilization of specialized hardware units.

Existing research has explored two largely separate directions to address these issues. One line of work focuses on optimizing Kyber for GPUs using careful memory management, kernel fusion, and improved traversal strategies for the Number Theoretic Transform (NTT), which is the dominant cost in polynomial multiplication. Another line of work investigates the use of AI accelerators—specifically NVIDIA Tensor Cores—to accelerate NTT by reformulating parts of the computation as matrix operations.

This report builds on these two complementary ideas. Motivated by their respective strengths and limitations, we study how kernel-level GPU optimizations and AI-accelerator-based computation can be combined to better exploit consumer-grade GPU hardware for post-quantum cryptography. The goal is not only to improve raw performance, but also to better understand how different GPU execution units can be used concurrently and efficiently for lattice-based cryptographic workloads.

1.2 OBJECTIVE

This project aims to design and implement an optimized solution for accelerating CRYSTALS-Kyber, a lattice-based post-quantum cryptosystem, using AI accelerators, specifically NVIDIA's Tensor Core. The primary objective is to address the computational challenges posed by polynomial multiplication in lattice-based cryptography by transforming it into a format that can be processed efficiently by Tensor Cores. This project seeks to improve the performance of CRYSTALS-Kyber without compromising its security by leveraging the specialized hardware of AI accelerators, traditionally used in machine learning, to speed up the most time-consuming operations in the cryptographic process.

The project will develop a custom "NTT box" that performs Number Theoretic Transforms (NTT) and its inverse (INTT) using Tensor Core's matrix multiplication capabilities. This solution will enable significant speedups in key cryptographic tasks such as key generation, encryption, and decryption, making it a viable solution for real-world post-quantum cryptography applications. Additionally, the project will explore how to balance precision and performance by experimenting with various data types like INT8 and FP16 to maximize the efficiency of Tensor Core operations.

The aim is to create a highly efficient, secure, and scalable implementation of CRYSTALS-Kyber that can serve as a reference for future cryptographic systems. By demonstrating the power of AI accelerators in accelerating cryptographic operations, this project seeks to contribute to the development of next-generation cryptography, specifically in the context of post-quantum algorithms. Ultimately, the project will provide an innovative solution for accelerating lattice-based cryptosystems, showcasing how AI hardware can be leveraged to meet the performance demands of modern cryptographic applications.

1.3 LITERATURE REVIEW

This section reviews prior work relevant to accelerating lattice-based post-quantum cryptography, with a focus on CRYSTALS-Kyber. The literature is grouped by platform and optimization strategy, and citations are numbered according to first appearance, following IEEE conventions. All cited works are drawn exclusively from the reference list provided in the project PPT.

1.3.1 FOUNDATIONS OF LATTICE-BASED CRYPTOGRAPHY AND KYBER

The modern development of post-quantum public-key encryption culminated in the selection of CRYSTALS-Kyber by NIST in 2022. Avanzi et al. describe the finalized Kyber design, its parameter sets, and security rationale based on the Module Learning With

Errors (MLWE) problem, along with improvements in noise handling and ciphertext compression [1]. This work establishes Kyber as the reference lattice-based KEM and motivates the need for efficient implementations across platforms.

At a systems level, Kyber's performance is dominated by polynomial arithmetic over the ring $R_q = Z_q[X]/(X^{256} + 1)$. The repeated use of the Number Theoretic Transform (NTT) in key generation, encryption, and decryption makes it the primary optimization target across all implementations.

1.3.2 CPU AND SIMD-BASED OPTIMIZATIONS

Initial performance improvements for Kyber focused on software optimization on CPUs. Hwang et al. demonstrate high-speed NTT implementations using AVX2 instructions, confirming that vectorization can significantly reduce latency for polynomial multiplication [2]. Similar ideas were explored on ARM platforms, where Nguyen et al. apply SIMD techniques to improve Kyber performance, although gains remain limited compared to massively parallel platforms [3]. Becker et al. further show that NEON SIMD instructions can accelerate Kyber on ARM CPUs, but these approaches are inherently restricted to specific instruction sets and processor families [4]. Collectively, CPU-based optimizations improve baseline performance but face hard limits due to restricted parallelism and memory bandwidth.

1.3.3 EMBEDDED AND ARM-FOCUSED IMPLEMENTATIONS

Several works target Kyber deployment on constrained devices. Alkim et al. present software optimizations for microcontrollers, improving efficiency but falling short of requirements for large scale or high-throughput applications [5]. Abdulrahman et al. extend this line of work by refining embedded optimization techniques for Cortex-M platforms, with a strong focus on low power consumption rather than raw speed [6].

Botros et al. optimize Kyber for ARM Cortex-M4 processors, demonstrating feasibility in embedded settings while highlighting the trade-off between performance and generality [7]. Sanal et al. further optimize Kyber for ARM64 systems using faster NTT routines and AES hardware acceleration, though these techniques remain hardware-specific [8]. While these works are essential for IoT and edge devices, their design constraints limit applicability to high-throughput environments.

1.3.4 GPU ACCELERATION USING CUDA CORES

GPUs offer large-scale parallelism and have therefore attracted attention for accelerating lattice-based cryptography. Gupta et al. present one of the earliest CUDA-based Kyber accelerators, optimizing subfunctions such as NTT and Keccak, but suffering from high global memory traffic [9]. Lee et al. explore fine-grained parallelization strategies for NTT

on RTX GPUs, demonstrating improved utilization but limited scalability across Kyber parameter sets [10]. Ji et al. significantly advance GPU-based Kyber acceleration through the HI-Kyber framework [11].

Their work introduces kernel fusion to eliminate unnecessary global memory accesses and proposes depth f irst NTT traversal strategies such as SLM, SDFS-NTT, and EDFS-NTT. These techniques dramatically reduce memory traffic and maximize register reuse, enabling million-level key exchange throughput on consumer GPUs. Despite these gains, HI-Kyber primarily utilizes CUDA cores and does not leverage specialized AI accelerators.

1.3.5 AI ACCELERATORS AND TENSOR-CORE-BASED APPROACHES

Recent GPUs incorporate AI accelerators, most notably Tensor Cores, which provide extremely high throughput for matrix multiply-and-accumulate operations. Wan et al. propose mapping Kyber's NTT computations to Tensor Cores using low-precision arithmetic and the WMMA programming model [12].

This approach demonstrates that Tensor Cores can be repurposed for cryptographic workloads, achieving substantial speedups for polynomial multiplication. An earlier framework by Wan et al. further generalizes this idea by exploring how lattice-based cryptographic workloads can be expressed for AI accelerators, though at the cost of complex data transformations and careful precision management [13].

These works show the potential of AI accelerators but focus primarily on isolated NTT acceleration rather than full-pipeline optimization. Complementary arithmetic optimizations are explored by Plantard, who proposes improved modular reduction techniques tailored for small moduli such as $q = 3329$, reducing the cost of modular multiplication in PQC schemes [14].

1.3.6 SUMMARY AND IDENTIFIED GAP

Across the literature, a clear progression is visible: CPU vectorization improves baseline performance, embedded implementations prioritize efficiency and power, GPU CUDA-core designs address parallelism and memory locality, and Tensor-Core approaches unlock new levels of arithmetic throughput. However, no single work fully integrates system-level GPU optimizations with AI accelerator-based computation.

CUDA-core-centric designs such as HI-Kyber excel at kernel fusion and memory efficiency, while Tensor Core-based designs offer unmatched performance for NTT. This separation motivates the approach explored in this report, which seeks to combine both strategies to better utilize heterogeneous GPU hardware for post-quantum cryptography.

Chapter 2

Proposed Methodology

2.1 METHODOLOGY DISCUSSION

This project presents a novel approach to accelerating lattice-based cryptography, specifically the CRYSTALS-Kyber post-quantum cryptosystem, using AI accelerators, namely NVIDIA's Tensor Cores. The aim is to transform the computationally intensive polynomial multiplication within Kyber into a form that can be processed efficiently by these AI-specific hardware components, traditionally optimized for machine learning tasks. The methodology focuses on leveraging the exceptional matrix multiplication capabilities of Tensor Cores to improve performance while ensuring the cryptographic operations retain their correctness. This transformation involves adapting the polynomial operations, particularly the Number Theoretic Transform (NTT), into a matrix multiplication format suitable for Tensor Cores, thereby achieving significant acceleration in cryptographic computations.

The core challenge is adapting the polynomial multiplication and NTT operations, typically designed for general-purpose processors, to fit the highly specialized and low-precision arithmetic of Tensor Cores. Tensor Cores, designed for matrix operations and low-precision computations, are ideally suited for the high parallelism required in polynomial multiplication, but transforming cryptographic operations into an AI-compatible format requires careful adjustments. The authors address this challenge by developing a custom "NTT box" that performs NTT and its inverse (INTT) in an efficient manner using CUDA C++ WMMA APIs. The proposed solution exploits multiple precision formats such as INT8 and FP16 to strike a balance between computational efficiency and precision.

The methodology also integrates precomputed twiddle factor tables, which help accelerate the NTT process by eliminating redundant calculations during the transformation. By organizing polynomial coefficients into matrices and utilizing Tensor Cores' matrix-multiply-and-add (MMA) capabilities, the method enables simultaneous processing of multiple polynomials in parallel. The NTT box utilizes precomputed tables for twiddle factors, which are loaded into memory to reduce latency, making the process more efficient. This design eliminates costly memory access delays, significantly improving the overall performance of polynomial multiplication in Kyber.

Empirical results show that the proposed method achieves substantial speedups, including a 6.47x improvement over the state-of-the-art implementation on the same GPU platform. Compared to the AVX2-based implementation, the Kyber-1024 parameter set shows speedups of 26x, 36x, and 35x in key generation, encryption, and decryption, respectively. These results underscore the potential of AI accelerators like Tensor Cores in accelerating post-quantum cryptography, particularly in lattice-based cryptographic schemes such as Kyber.

The methodology also addresses key security considerations, particularly against side-

channel attacks. The use of Tensor Cores, which are not susceptible to traditional timing attacks due to their parallel execution model, mitigates risks associated with side-channel vulnerabilities. The system also incorporates defense mechanisms against timing leaks by eliminating secret-dependent conditional branches in the CUDA kernels, further enhancing the security of the cryptographic implementation.

This project is implemented in four stages. The first phase involves adapting CRYSTALS-Kyber to leverage Tensor Cores for polynomial multiplication. In the second phase, the integration of precomputed tables and memory optimizations is introduced to further enhance performance. The third phase involves testing the solution with larger parameter sets, such as Kyber-768 and Kyber-1024, to evaluate scalability. The final phase refines the system by optimizing memory access patterns and exploring further use of AI accelerators for other cryptographic primitives. This phased approach allows for a systematic evaluation of the proposed methods and ensures that each step contributes effectively to the overall performance improvement.

2.1.1 PHASE 1: INITIAL CRYSTALS-KYBER ACCELERATION WITH TENSOR CORES

The first step in this methodology is to implement a baseline solution by adapting CRYSTALS-Kyber to use NVIDIA's Tensor Core for accelerating polynomial multiplication. The primary goal in this phase is to establish a functional implementation that uses Tensor Cores' matrix multiplication capabilities to replace traditional CPU-based operations. The first task involves converting the polynomial multiplication process into a matrix format that can be efficiently processed by the Tensor Core architecture. This transformation is done by adapting the NTT computation, traditionally used in lattice-based cryptography, into matrix multiplications compatible with Tensor Cores.

To do this, the polynomials involved in Kyber's cryptographic operations are organized into matrices, and the twiddle factors necessary for the NTT operation are precomputed and stored. The core component of this phase is the development of the NTT box, a custom module that implements the NTT and INTT operations using the tensor-specific matrix multiplication and accumulation features of the Tensor Cores. The results of this phase will serve as a foundation for the subsequent phases, providing a working version of the system that demonstrates the feasibility of using Tensor Cores for lattice-based cryptography.

2.1.2 PHASE 2: INTEGRATING PRECOMPUTED TABLES AND OPTIMIZING MEMORY ACCESS

In the second phase, the focus shifts to optimizing the system by incorporating precomputed twiddle factor tables and fine-tuning memory access patterns. The precomputed tables allow for the fast retrieval of twiddle factors during the NTT operation, eliminating the need for repeated calculations. This reduces the overall computation time and enhances the performance of the system. Additionally, optimizations are made to ensure that the memory access patterns are aligned with Tensor Core's capabilities, minimizing memory latency and ensuring that the polynomial data is efficiently loaded into the accelerator.

A key part of this phase is the integration of specialized memory management techniques that ensure data is organized and accessed in a way that maximizes throughput. This phase also includes profiling and benchmarking to evaluate the performance gains achieved through these optimizations. By the end of this phase, the system will demonstrate a fully optimized pipeline that can process larger parameter sets more efficiently.

2.1.3 PHASE 3: SCALABILITY TESTING WITH LARGER PARAMETERS

The third phase focuses on testing the scalability of the implementation with larger parameter sets such as Kyber-768 and Kyber-1024. In this phase, the system is evaluated under different cryptographic workloads to understand how well it handles increasing computational complexity. The goal is to verify whether the optimizations made in the previous phases hold up as the size of the cryptographic tasks grows.

Scalability tests will involve measuring the performance of key operations like key generation, encryption, and decryption across a range of parameter sizes. The results will help assess the effectiveness of the proposed solution in real-world cryptographic scenarios, where larger parameters are required for higher security levels. Any performance bottlenecks identified during this phase will be addressed in the next iteration of optimization.

2.1.4 PHASE 4: FINAL OPTIMIZATIONS AND SECURITY ENHANCEMENTS

The final phase will refine the implementation by further optimizing memory access, reducing latency, and integrating advanced security measures. This phase will involve enhancing the security of the implementation by incorporating defenses against side-channel attacks, such as timing attacks, by ensuring that the execution paths are constant-time and independent of sensitive data. Additionally, further optimizations will be made to improve the throughput and minimize any remaining inefficiencies in the system.

In this phase, the system will also undergo comprehensive testing to validate its robustness, performance, and security. The final deliverable will be a highly efficient, secure, and scalable cryptographic implementation that demonstrates the power of AI accelerators in accelerating post-quantum cryptography.

2.2 PROPOSED ARCHITECTURE DIAGRAM

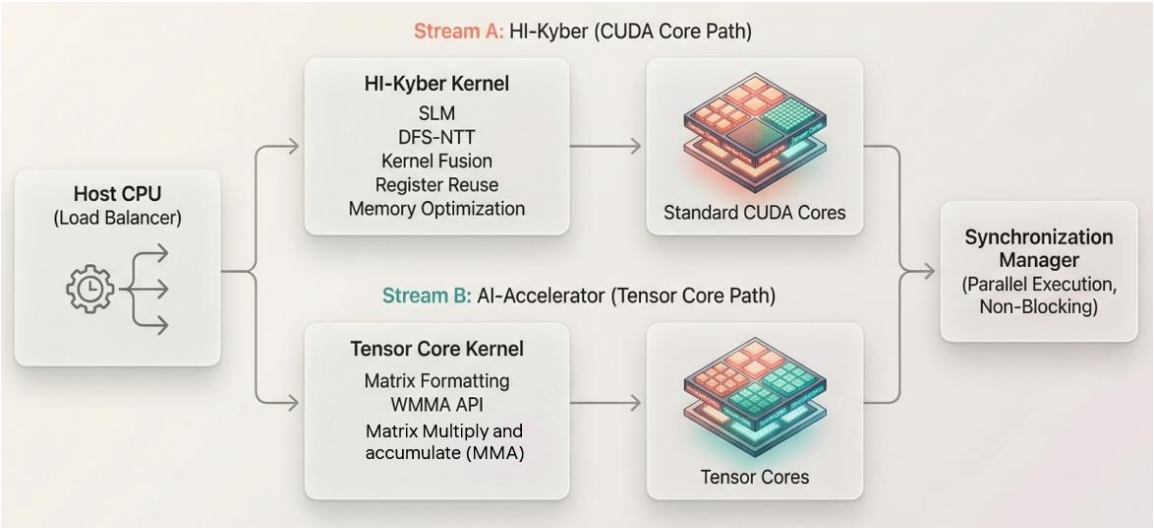

Figure 1(a) High level system architecture